

Analyzing Political Discourse and Toxicity on Reddit and 4chan

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1 ABSTRACT

This paper presents an analysis of the political climate on popular social media platforms, 4chan and Reddit, and the political trends, opinions, and discussions each fringe online community exhibits. The study uses user sentiment analysis and post frequency data to determine the type of political conversations these user-generated content platforms discuss. The data used in the analysis spans from November 5th to November 13th. We augment the data with distinct labels such as toxicity scores, the confidence of these scores, and timestamps. We present a statistical analysis of the dataset, shedding light on the most popular political communities on Reddit and when each community was most active. Our dataset also provides insight into the patterns of user behavior and evolution of political discourse over time.

1.1 Introduction

Modern mainstream media typically does not capture the full spectrum of individual sentiments and perspectives that emerge in online communities, often leaving out nuanced, raw, and unfiltered discussions. This gap underscores the importance of studying platforms like Reddit and 4chan, where diverse voices are directly expressed by users, offering a more comprehensive view of public sentiment.

Reddit and 4chan are two of the most popular online political communities, where users engage in discussions, debates, and exchange of ideas on a wide range of political topics. These platforms serve as social platforms, where grassroots conversations and formation of ideological echo chambers form. While Reddit offers a wide variety of subreddits catering to different political leanings and conversations with varying levels of moderation, 4chan's politically incorrect board (/pol/) operates as an anonymous, minimally operated space. And, with the ephemeral nature of its posts, fosters unfiltered and polarizing discourse among its users.

Overall, 4chan is widely known for its large amount of users, anonymity, and lack of moderation. This creates an ideal community for users to post unfiltered posts and sometimes harmful content. Despite its rapid spread of conspiracy theories, hatespeech, and extreme ideologies, it allows for niche discussions and cultural phenomena that would not be allowed in more moderated mediums. This unique dynamic makes 4chan a significant component of our dataset, offering valuable insights into unfiltered discourse.

To perform an analyses on the correlation of sentiment data with online political communities, we used Flask Web API to serve as the backend framework for processing and analyzing data in real time. The Flask Web API allows users to input parameters such as which subreddit or board they would like to view. Unfortunately, due to time constraints we could not add as many dynamic features as we had planned.

1.2 Research Question

The research question we had sought out to answer from our proposal was to identify toxic posts in online communities and also analyze language patterns to determine sentiment, intent, and potential indicators of harmful discourse.

1.3 Related Work

Research into online political discourse and sentiment has been prevalent for the past decade, particularly on Reddit and 4chan, where user-generated data is readily available. In Hofmann et al. 2022[1], the study focuses on the Reddit Politosphere, a large-scale resource of online political conversation of more than 600 subreddits. By using subreddit metadata and analyzing text using natural language processing, sentiment data could be gathered from posts, comments, and upvotes or downvotes. These studies highlight the importance of leveraging user-generated content to understand political sentiment trends and community dynamics in online platforms.

2 METHODOLOGY AND DATASET

2.1 Data Constraints

The data collection for this paper spanned from November 5th to November 13th, and though we had relevant data prior, our Reddit and 4chan data, it was compromised before we could fully integrate it into the analysis.

We focused on the collection of data from political subreddits and /pol/ on 4chan to ensure a diverse dataset, targeting r/politics, r/NeutralPolitics, and r/PoliticalDiscussion. During this period, we collected not only political data, but also relevant current events that significantly influenced the nature of discussions on these platforms. These events included suspect charged in the deadly shooting of

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Rank	Word	Frequency
1	Trump	755
2	Rules	449
3	Syria	429
4	Media	419
5	Support	391
6	Right	359
7	Civil	313
8	War	311
9	Government	309
10	Violence	289

Table 1: Most Frequently Used Political Words on Reddit

Rank	Word	Frequency
1	Right	446
2	Women	397
3	White	353
4	Jews	353
5	Israel	190
6	Russian	181
7	Kill	142
8	Jewish	140
9	America	136
10	Trump	118

Table 2: Most Frequently Used Words on 4chan /pol/

the United Healthcare CEO, Brian Thompson and president-elect Donald Trump, who is soon to be inaugurated.

2.2 Data Collection

Our data was collected and stored into PostgreSQL, where we would perform queries for indexing, JSON data, and implemented with the Moderate Hatespeech API to analyze toxicity of posts directly from the database. We fetched post titles and ids, timestamps, toxicity, confidence of toxicity, and the JSON metadata. We also collected the comments to each post or thread, for more in-depth analysis of the conversations and interactions within each community.

2.3 Tables

Table 1 and Table 2 illustrate the most political or politically adjacent words most frequently used across the collected datasets. On Reddit, Trump is the most frequently used word, followed by terms like "rules", "Syria", and "media", reflecting a focus on mainstream political figures and places. These words highlight the structure and topical nature on Reddit. In contrast, the most used words on 4chan is "right", followed by "women", "white", and "Jews", reflecting a starkly different focus in discussions compared to Reddit. These words suggest that conversations on 4chan's /pol/ often revolve around identity politics, social hierarchies, and polarizing ideological themes.

3 FLASK

We use Flask as our primary Web API to facilitate the seamless interaction between the backend processing and client interface. Flask's

Post Visualization by Subreddits



Figure 1: Visualization: /visualize/subreddits - Post Frequency of Different Political Subreddits

modular design allows us to define flexible endpoints for data analysis, querying, and visualization. Users can provide parameters such as subreddit or board selection, allowing users to customize data retrieval and analysis.

3.1 Post Frequency Table

This figure displays the frequency distribution of posts across various political subreddits visualized as a bar graph. This shows that r/politics is by far the most popular subreddit, showing a higher post frequency compared to more niche communities like r/NeutralPolitics or r/PoliticalDebate. These smaller subreddits may

Post Trends in Different Subreddits



Figure 2: Visualization: /visualize/dates - Temporal Post Frequency on Reddit

have more specialized discussions, providing valuable data on more niche political topics or specific ideologies than more common, mainstream views.

This figure displays the temporal post frequency of both Reddit and 4chan, which can be dynamically viewed based on user input. As mentioned, data is lacking prior to November 25th due to data loss. Despite this, we collected relevant data following, including the incident involving the United Healthcare CEO, Brian Thompson, which can be seen in the sudden surge in posts on December 6th. As expected, the greatest surge in posts came from r/politics, where many of the mainstream current events are discussed by users. During this time, we saw no increase in toxic posts on any subreddits, and only one toxic post was collected on December 7th of a post about gender politics involving Donald Trump.

3.2 Analysis

On 4chan, we actually collected a much larger amount of toxic posts, totaling 201 posts. However, many of the posts were consistent throughout this event, so we don't expect this to be an abnormality on 4chan's /pol/ board. Instead, we found that many of the toxic posts involved racism, gender discrimination, and other forms of discrimination. In contrast to Reddit, 4chan demonstrates a consistent pattern of toxicity in its posts, which is expected due to its anonymous and unmoderated nature. Based on our data, 4chan's constant presence of toxic posts was not affected by any largescale current events, instead, it seems to be a embedded in the platform's discourse in general. This consistent nature of toxicity is fueled by the freedom of its users, which suggest an echo chamber effect. In this case of 4chan, this results in a culture where racism, sexism, and misogyny are not only tolerated but normalized. The users who participate in these discussions often validate each other's toxic behaviors, furthering these harmful ideologies in this community.

4 CONCLUSION

4.1 Limitations

Due to data loss prior to November 5th, we lost relevant data pertaining to the presidential election, collecting most of the events afterwards, but not the user sentiment during or prior. During this

time, we expected user engagement to be at its peak, as the presidential election was expected to drive intense discussion across online political communities. The loss of this data limited our ability to analyze sentiment shifts and user behavior during critical moments of the election cycle, such as debates, election night, and immediate post-election reactions.

5 REFERENCES

[1] Hofmann, V., Schütze, H., & Pierrehumbert, J. B. (2022). The Reddit Politosphere: A Large-Scale Text and Network Resource of Online Political Discourse.